Assignement

2023-04-06

Question 1

Structure of the "flights" dataset

```
## tibble [336,776 x 19] (S3: tbl_df/tbl/data.frame)
##  $ year          : int [1:336776] 2013 2013 2013 2013 2013 2013 2013 2013 2013 2013 ...
##  $ month         : int [1:336776] 1 1 1 1 1 1 1 1 1 1 ...
##  $ day           : int [1:336776] 1 1 1 1 1 1 1 1 1 1 ...
##  $ dep_time      : int [1:336776] 517 533 542 544 554 554 555 557 557 558 ...
##  $ sched_dep_time: int [1:336776] 515 529 540 545 600 558 600 600 600 600 ...
##  $ dep_delay     : num [1:336776] 2 4 2 -1 -6 -4 -5 -3 -3 -2 ...
##  $ arr_time      : int [1:336776] 830 850 923 1004 812 740 913 709 838 753 ...
##  $ sched_arr_time: int [1:336776] 819 830 850 1022 837 728 854 723 846 745 ...
##  $ arr_delay     : num [1:336776] 11 20 33 -18 -25 12 19 -14 -8 8 ...
##  $ carrier       : chr [1:336776] "UA" "UA" "AA" "B6" ...
##  $ flight        : int [1:336776] 1545 1714 1141 725 461 1696 507 5708 79 301 ...
##  $ tailnum       : chr [1:336776] "N14228" "N24211" "N619AA" "N804JB" ...
##  $ origin        : chr [1:336776] "EWR" "LGA" "JFK" "JFK" ...
##  $ dest          : chr [1:336776] "IAH" "IAH" "MIA" "BQN" ...
##  $ air_time      : num [1:336776] 227 227 160 183 116 150 158 53 140 138 ...
##  $ distance      : num [1:336776] 1400 1416 1089 1576 762 ...
##  $ hour          : num [1:336776] 5 5 5 5 6 5 6 6 6 6 ...
##  $ minute        : num [1:336776] 15 29 40 45 0 58 0 0 0 0 ...
##  $ time_hour     : POSIXct[1:336776], format: "2013-01-01 05:00:00" "2013-01-01 05:00:00" ...
```

The dataset shape are 336.776 $\times$ 19 (336.776 observations and 19 features)

Filtering

```
## [1] 842
```

The filtered dataset by departed 1st January contain 842 rows.

Question 2

Question 3

Question 4

```
## # A tibble: 58,368 x 5
##     year month   day arr_delay arr_delay_category
##    <int> <int> <int>     <dbl> <chr>
##  1  2013     1     1        -2 On-time
##  2  2013     1     1        -3 On-time
##  3  2013     1     1        -4 On-time
```

```
##  4  2013     1     1         4 On-time
##  5  2013     1     1         3 On-time
##  6  2013     1     1         5 On-time
##  7  2013     1     1         1 On-time
##  8  2013     1     1         0 On-time
##  9  2013     1     1        -3 On-time
## 10  2013     1     1        -5 On-time
## # ... with 58,358 more rows
```

```
## # A tibble: 252,173 x 5
##     year month   day arr_delay arr_delay_category
##    <int> <int> <int>     <dbl> <chr>
##  1  2013     1     1        11 Common delays
##  2  2013     1     1        20 Common delays
##  3  2013     1     1        33 Common delays
##  4  2013     1     1       -18 Common delays
##  5  2013     1     1       -25 Common delays
##  6  2013     1     1        12 Common delays
##  7  2013     1     1        19 Common delays
##  8  2013     1     1       -14 Common delays
##  9  2013     1     1        -8 Common delays
## 10  2013     1     1         8 Common delays
## # ... with 252,163 more rows
```

```
## # A tibble: 26,235 x 5
##     year month   day arr_delay arr_delay_category
##    <int> <int> <int>     <dbl> <chr>
##  1  2013     1     1       137 Long delay
##  2  2013     1     1       851 Long delay
##  3  2013     1     1       123 Long delay
##  4  2013     1     1       145 Long delay
##  5  2013     1     1        93 Long delay
##  6  2013     1     1       103 Long delay
##  7  2013     1     1       127 Long delay
##  8  2013     1     1       125 Long delay
##  9  2013     1     1        NA Long delay
## 10  2013     1     1        NA Long delay
## # ... with 26,225 more rows
```

Question 5

The delays in the descending order of the "mean_arr_delay"

```
## # A tibble: 105 x 3
##    dest  mean_arr_delay mean_dep_delay
##    <chr>          <dbl>          <dbl>
##  1 CAE             41.8           35.6
##  2 TUL             33.7           34.9
##  3 OKC             30.6           30.6
##  4 JAC             28.1           26.5
##  5 TYS             24.1           28.5
##  6 MSN             20.2           23.6
##  7 RIC             20.1           23.6
##  8 CAK             19.7           20.8
##  9 DSM             19.0           26.2
```

```
## 10 GRR             18.2           19.5
## # ... with 95 more rows
```

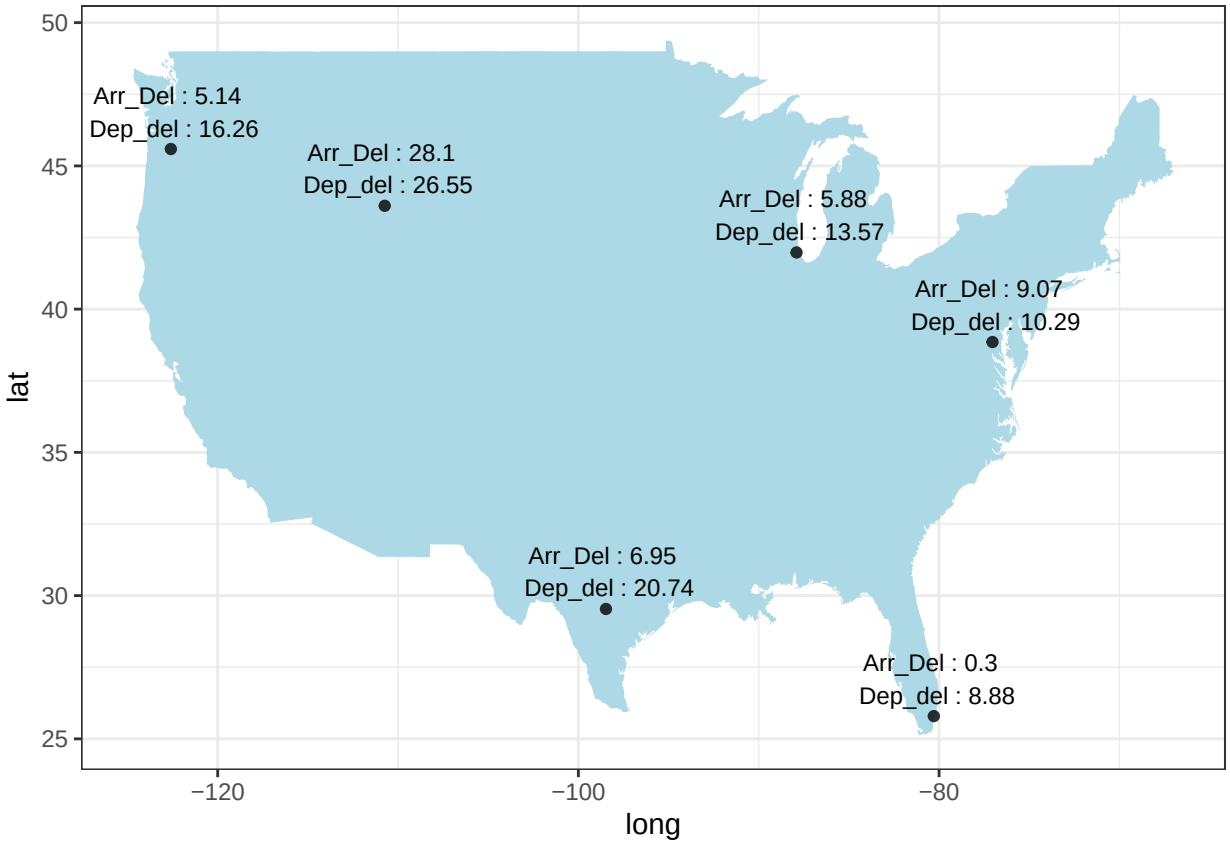
The delays in the descending order of the "mean_dep_delay"

```
## # A tibble: 105 x 3
##    dest  mean_arr_delay mean_dep_delay
##    <chr>          <dbl>          <dbl>
##  1 CAE             41.8           35.6
##  2 TUL             33.7           34.9
##  3 OKC             30.6           30.6
##  4 BHM             16.9           29.7
##  5 TYS             24.1           28.5
##  6 JAC             28.1           26.5
##  7 DSM             19.0           26.2
##  8 RIC             20.1           23.6
##  9 ALB             14.4           23.6
## 10 MSN             20.2           23.6
## # ... with 95 more rows
```

Question 6

```
## # A tibble: 6 x 6
##   dest  mean_arr_delay mean_dep_delay name                            lat    lon
##   <chr>          <dbl>          <dbl> <chr>                         <dbl>  <dbl>
## 1 DCA            9.07           10.3  Ronald Reagan Washington Natl  38.9  -77.0
## 2 JAC           28.1            26.5  Jackson Hole Airport           43.6 -111.
## 3 MIA            0.299           8.88 Miami Intl                     25.8  -80.3
## 4 ORD            5.88           13.6  Chicago Ohare Intl             42.0  -87.9
## 5 PDX            5.14           16.3  Portland Intl                  45.6 -123.
## 6 SAT            6.95           20.7  San Antonio Intl               29.5  -98.5
```

Question 7



Question 8

```
## # A tibble: 12 x 5
##    month mean_arr_delay mean_dep_delay cumulative_arr_delays cumulative_dep_de~1
##    <int>          <dbl>          <dbl>                <dbl>                <dbl>
##  1     1          6.13           10.0                  6.13                 10.0
##  2     2          5.61           10.8                 11.7                  20.9
##  3     3          5.81           13.2                 17.6                  34.1
##  4     4         11.2            13.9                 28.7                  48.0
##  5     5          3.52           13.0                 32.2                  61.0
##  6     6         16.5            20.8                 48.7                  81.9
##  7     7         16.7            21.7                 65.4                 104.
##  8     8          6.04           12.6                 71.5                 116.
##  9     9         -4.02            6.72                67.5                 123.
## 10    10         -0.167           6.24                67.3                 129.
## 11    11          0.461           5.44                67.8                 135.
## 12    12         14.9            16.6                 82.6                 151.
## # ... with abbreviated variable name 1: cumulative_dep_delays
```

Question 9

```
## # A tibble: 24 x 3
##    month delay_type           delays
```

```
##    <int> <chr>                 <dbl>
##  1     1 cumulative_arr_delays   6.13
##  2     1 cumulative_dep_delays  10.0
##  3     2 cumulative_arr_delays  11.7
##  4     2 cumulative_dep_delays  20.9
##  5     3 cumulative_arr_delays  17.6
##  6     3 cumulative_dep_delays  34.1
##  7     4 cumulative_arr_delays  28.7
##  8     4 cumulative_dep_delays  48.0
##  9     5 cumulative_arr_delays  32.2
## 10     5 cumulative_dep_delays  61.0
## # ... with 14 more rows
```

